Haemoglobinopathies are a set of diseases caused by synthesis of structurally abnormal haemoglobins, insufficient amount of haemoglobins or both. Sickle cell anemia, thalasemia, porphyria etc are examples of haemoglobinopathies.

3.6 Collagen - an example for fibrous protein

Collagen is the most abundant protein found in humans. Unlike globular structures discussed above, collagen forms a long coiled fibrous structure. Each collagen molecule consists of three polypeptide chains which forms an elongated triple helical structure as described in figure 3.16

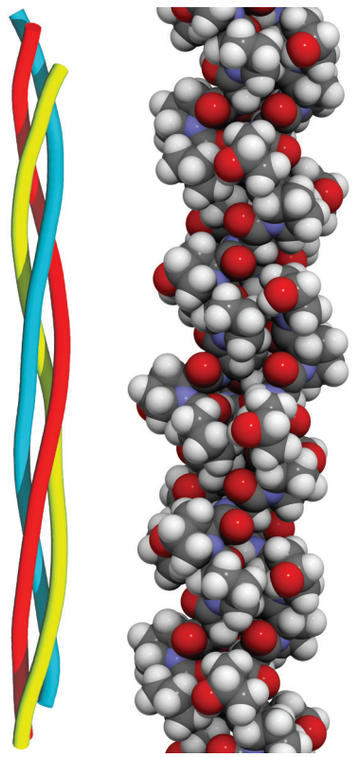


Figure 3.16 Structure of Collagen

The amino acid sequence of these polypeptide chains are always repeating units of Gly-X-Y. Where X is often proline and Y is either hydroxyproline or hydroxylysine. The hydroxyl group of hydroxylysine can also be glycosylated with glucose or galactose.

There are various types of collagen which can be broadly categorized into three groups.

a. The fibril forming collagens present in skin, bone, cartilages, tendons and blood vessels etc provide tensile strength to the corresponding tissues.

b. The network forming collagens form network like structures beneath the membranes providing them mechanical strength.

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